Final Project

Note: All files and information related to the final project are located in the various folders starting with "Final Project" prefix.

The aim of this Final Project is to practically implement the ideas from the course, specifically related to Levy Probability Density Function. You will be given the recent since inception 1-minute price history of Day Trading Session of the following markets:

Symbol	Name	Region	Exchange	History	Day Session	File Name
YM	Dow Jones E-mini	America/Chicago	CBOT/CME	5/1/2002-1/14/2026	8:30 AM – 3:15 PM	1min-Day-YM-skip.csv
NQ	NASDAQ 100 E-mini	America/Chicago	CME	7/1/1999-1/14/2026	8:30 AM – 3:15 PM	1min-Day-NQ-skip.csv
ER	Russell 2000 E-mini	America/New York	CBOT/CME	1/2/2002-1/14/2026	8:30 AM – 3:15 PM	1min-Day-ER-skip.csv
ES	S&P 500 E-mini	America/Chicago	CME	9/10/1997-1/14/2026	8:30 AM – 3:15 PM	1min-Day-ES-skip.csv
PT	S&P Canada 60	America/Montreal	Montreal Stock Exchange	10/1/1999-1/14/2026	9:30 AM – 4:15 PM	1min-Day-PT-skip.csv
FT	FTSE 100	Europe/London	ICE Futures Exchange	7/1/1998-1/14/2026	8:00 AM – 4:30 PM	1min-Day-FT-skip.csv
DA	DAX	Europe/Berlin	EUREX	1/2/1997-1/14/2026	9:00 AM – 5:30 PM	1min-Day-DA-skip.csv
CF	CAC-40	Europe/Paris	Euronext Derivatives Exchange	5/2/2000-1/14/2026	9:00 AM – 5:30 PM	1min-Day-CF-skip.csv

This data is created by a software TickData, which is owned by OneTick data provider. The data files were created with an option "Skip", which means if there is no transactions with prices and volume for a given minute, then this minute is simply skipped in the file.

You will need the following theoretical results, which were derived in the theoretical sessions. The expression for the Levy distribution density function can we written in a more universal form as follows:

$$P_{\alpha,\gamma}(x,\tau) = (\gamma\tau)^{-\frac{1}{\alpha}} F_\alpha(\tilde{x}),$$

where

$$F_\alpha(\tilde{x}) = \frac{1}{\pi}\int_0^{+\infty} e^{-\tilde{q}^\alpha}\cos(\tilde{q}\tilde{x})d\tilde{q}, \text{ and } \tilde{x} = \frac{x}{(\gamma\tau)^{\frac{1}{\alpha}}}, \tilde{q} = q(\gamma\tau)^{\frac{1}{\alpha}}.$$

The leading-order term of the Taylor series of the Levy distribution density function around $x = 0$ is:

$$P_{\alpha,\gamma}(0,\tau) = \frac{(\gamma\tau)^{-\frac{1}{\alpha}}}{\pi\alpha}\Gamma\left(\frac{1}{\alpha}\right).$$

The leading-order term in the asymptotical series of the Levy distribution density function for $x \to \infty$ is:

$$P_{\alpha,\gamma}(x,\tau) \sim \frac{\gamma\tau}{\pi}\sin\left(\frac{\pi\alpha}{2}\right)\Gamma(\alpha+1)\frac{1}{|x|^{\alpha+1}}.$$

To justify the use of 1-minute resolution price data, consider the following data measured from the above markets:

	YM	NQ	ER	ES	PT	FT	DA	CF
$\boldsymbol{\sigma_1}$	6.15	2.33	0.51	0.68	2.78	3.48	2.61	0.46
$\boldsymbol{Ask - Bid}$	1.30	0.48	0.22	0.25	2.50	1.20	2.00	0.22
$\boldsymbol{\frac{\sigma_1}{Ask - Bid}}$	4.73	4.90	2.32	2.71	1.11	2.90	1.31	2.09

Here we denoted as σ_1 the standard deviation of 1-minute price changes and $Ask - Bid$ is the average bid-offer spread.

As one can see from this data, all values of standard deviations are more than one bid-offer spread. For values of this ratio below one, a substantial portion of observed price changes reflects the so-called "bid-ask bounce" phenomenon, when the successive trades occur alternatingly at bid and ask prices without any true movement of the underlying price.

Measuring the Levy exponent is highly sensitive to the details of the details of the measurement technique, which we will describe in detail now.

As we anticipate to obtain the algebraic scaling law, we will use the equidistant grid in logarithmic time separation τ. To estimate the error bounds of the scaling exponents, the two choices of τ-grid will be used. The first one with the size $N_\tau = 20$ is shown in the following table:

$\boldsymbol{i}$	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
$\boldsymbol{\tau_i}$	1	2	3	6	10	18	32	56	100	178
$\boldsymbol{i}$	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
$\boldsymbol{\tau_i}$	316	562	1000	1778	3162	5623	10000	17783	31623	56234

The second one with the size $N_\tau = 37$ is shown in the following table:

$\boldsymbol{i}$	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
$\boldsymbol{\tau_i}$	1	2	3	4	6	7	10	13	18	24
$\boldsymbol{i}$	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
$\boldsymbol{\tau_i}$	32	42	56	75	100	133	178	237	316	422
$\boldsymbol{i}$	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
$\boldsymbol{\tau_i}$	562	750	1000	1334	1778	2371	3162	4217	5623	7499
$\boldsymbol{i}$	**31**	**32**	**33**	**34**	**35**	**36**	**37**			
$\boldsymbol{\tau_i}$	10000	13335	17783	23714	31623	42170	56234			

As one can see, we chose the largest time separation $\tau = 56234$, which for the S&P 500 E-mini market (which has 405 minutes in a Day trading session) corresponds to around 6 months, which is more than sufficient, as we expect convergence to a Gaussian occur at scales of around one month.

The next step is for a given number of bins along the x-axis (or price-change) N_x, to determine such value of $x_{max} > 0$, which defines the range of values of price changes $-x_{max} \le x \le x_{max}$, within which we will be counting the price changes in each bin. We believe that such a procedure should rely on the actual price volatility of the smallest $\tau = 1$ minute fluctuations σ_1, measured above. We will use the following phenomenological formula to estimate x_{max}:

$$x_{max} = \Delta N_x \times max\left\{\left[\frac{\varepsilon\sigma_1}{\Delta}\right], 1\right\},$$

with a small threshold parameter $0.1 \le \varepsilon \le 0.5$, Δ is the average bid-ask spread, and $[\cdot]$ means averaging to the nearest integer. You need to use the number of bins $N_x = 2000$.

The procedure for estimating the Levy tail exponent α consists of the following steps:

1. For a specific threshold $\varepsilon \in \{0.1,0.2,0.3,0.4,0.5\}$ and a specific number of time shifts $N_\tau \in \{20,37\}$, calculate the corresponding x_{max} using the formula above.
2. For all time shifts τ_i corresponding to a given set $N_\tau \in \{20,37\}$ defined above, and for all bins $-x_{max} \leq x \leq x_{max}$, calculate the PDF $P_{\alpha,\gamma}(\tau_i, x_i)$.
3. Using the linear regression and the leading-order Taulor series behavior above, estimate the Levy exponent α and its corresponding R^2 value for controlling the sample error.
4. Using the estimated average value of Levy exponent α, show the universal function $F_\alpha(\tilde{x})$ for a wide range of τ values. If a reasonably good collapse for various values of τ is observed, the Levy distribution is indeed describing the price data. Here you can additionally symmetrize the $F_\alpha(\tilde{x})$, which means use the additional averaging $\frac{F_\alpha(-\tilde{x})+F_\alpha(\tilde{x})}{2}$.

The tool that I propose you use for the PDF calculating is the C-language code that the Instructor has written and debugged for you: ***pdf-1min.c***. This course code is written for Linux OS, you compile it using the following command: "***gcc pdf-1min.c -o pdf-1min.exe -lm***", and you run is using the following command: "***./pdf-1min.exe***". It produces the following output files: "***ProbabilityDensity.csv***", "***FilledTimePrice.csv***", "***Frequencies.csv***", of which the most important is the file "***ProbabilityDensity.csv***", which contains the PDF measurements that you need. The executable needs the data file, such as "***1min-Day-ES-skip.csv***". The control is done using commenting and uncommenting corresponding pieces in the #define part of the file ***pdf-1min.c***. For example, for the ES market your uncommented #define part should look like this:

```
#define FILENAME "1min-Day-ES-skip.csv"
#define MAXSIZE 2895667
#define XMAX 50000
#define XMIN -50000
#define TICK 0.25
#define OPENHOUR 8
#define OPENMINUTE 30
#define MULTIPLIER 100.0
```

All other similar #define parts should be commented out. XMAX is define by a formula above.

Using this procedure, calculate the Levy exponents for all the markets, all values of ε for every value of N_τ, and fill out the following table:

$\boldsymbol{\varepsilon}$	**YM**	**NQ**	**ER**	**ES**	**PT**	**FT**	**DA**	**CF**
0.1								
0.2								
0.3								
0.4								
0.5								

Estimate the sample error of your measurements with each N_τ and for both N_τ together. Compare the values of Levy exponents for different markets. Try to make some inferences and conclusions from your experimental results.

This Final Project in an open-book which means that you can and should use the Instructor's handouts and the corresponding reading material, as well as any additional materials provided to you. Instructor has performed all these calculations and will be able to compare your to his. If your calculations are done correctly, you and he should be able to match the results with sufficient accuracy.

You are expected to write the details of your experimental measurements and the results of your analysis in the Final Project paper. You are given two weeks to write the Final Project paper. I encourage you not to delay starting the work as workload is meant for several days of work and not as a one-night effort.

Do not hesitate to ask the Lecturer any questions related to this.

Good luck!